CP 160 - Web Programming and Design

Lab 6: More CSS

Hands-On Practice 1: Configure a logo header area

You'll configure a logo header area that uses a background image and rounded borders in this Hands On Practice.

1. Create a new folder called **bistroch6**. Copy the lighthouselogo.jpg , background.jpg files and **starter2.html** file in the **starters** folder to your bistroch6 folder. Launch a browser to display the starter2.html web page shown in **Figure 1**.

2. Launch a text editor and open the starter2.html file. Save the file as **index.html**. Edit the embedded CSS and code an h1 element selector with style declarations that will configure the lighthouselogo.jpg image as a background image that does not repeat: height of 100px, width of 700px, font size of 3em, 150px of left padding, 30px of top padding, and a border radius of 15px. The style declarations follow:

```
h1 {    background-image: url(lighthouselogo.jpg);
        background-repeat: no-repeat;
        height: 100px; width: 700px;
        font-size: 3em;
        padding-left: 150px;
        padding-top: 30px;
        border-radius: 15px; }
```

3. Save the file. When you test your index.html file in a browser, it should look similar to the one shown in **Figure 2.**

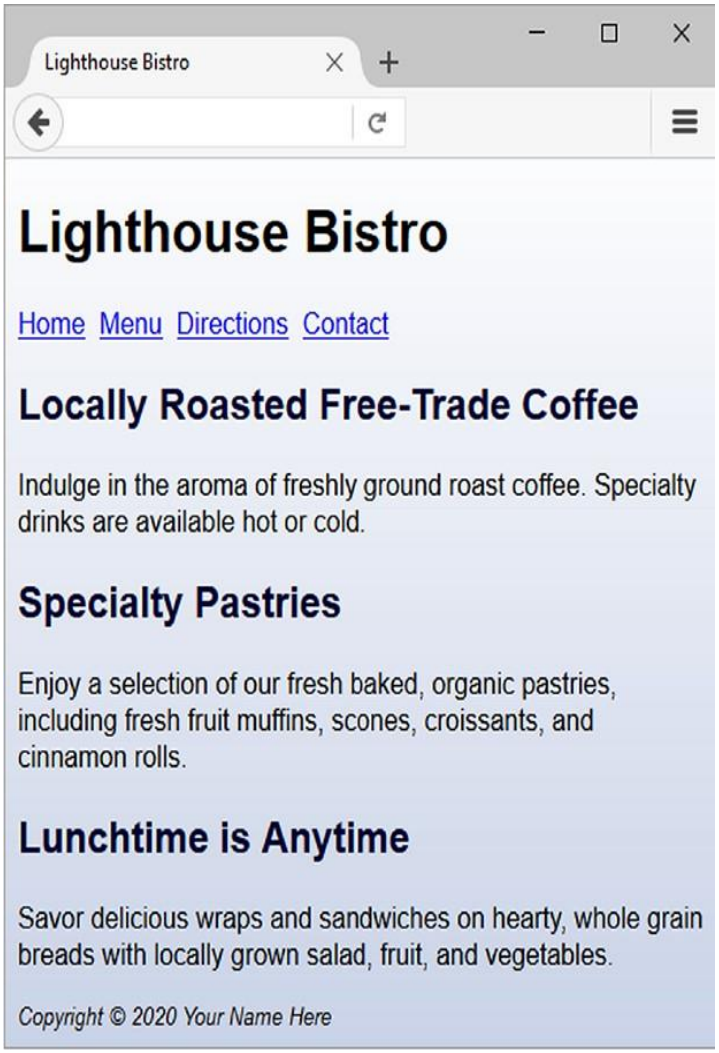

Figure 1 The starter2.html file.

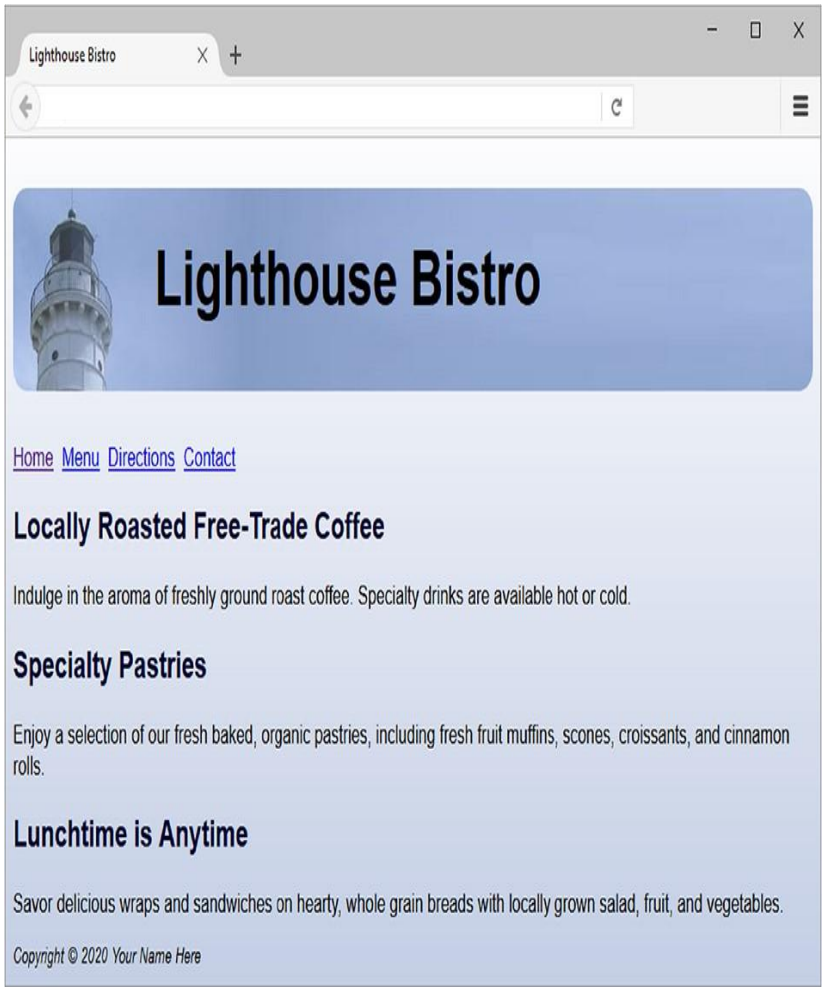

Figure 2 The web page with the logo area configured.

Hands-On Practice 2: Center a web page

You will practice modifying and centering a web page in this Hands-On Practice as you update the index.html file from **Hands-On Practice 1** (shown in **Figure 2**). A common design practice is to configure the background color of the wrapper or container to be a light, neutral color that provides good contrast with text. When complete, your web page will be similar to the one shown in **Figure 3**.

Create a new folder called **centerch6**. Copy the index.html, background.jpg, and lighthouselogo.jpg files from **bistroch6** to centerch6.

Launch a text editor and open the index.html file.

1. Edit the embedded CSS and configure the h1 selector. Remove the width style declaration. Configure a medium blue (#9DB3DC) background color and a 0 top margin (use the `margin-top` property).

```
h1 { background-image: url(lighthouselogo.jpg);
     background-color: #9DB3DC;
     background-repeat: no-repeat;
     height: 100px; font-size: 3em;
     padding-left: 150px; padding-top: 30px;
     border-radius: 15px;
     margin-top: 0;
}
```

2. Edit the embedded CSS and configure a new selector, an id named `container`. Add style declarations for the `background-color`, `padding`, `width`, `min-width`, `margin-left`, and `margin-right` properties as follows:

```
#container { background-color: #FFFFFF;
             padding: 2em;
             margin-left: auto;
             margin-right: auto;
             width: 80%;
             min-width: 800px; }
```

3. Edit the HTML. Configure a div element assigned to the id `container` that "wraps" or contains the code within the body section. Code an opening div tag on a new line after the opening body tag. Assign the div to the id named `container`. Code the closing div tag on a new line before the closing body tag.

Save the file. When you test your index.html file in a browser, it should look similar to **Figure 3**.

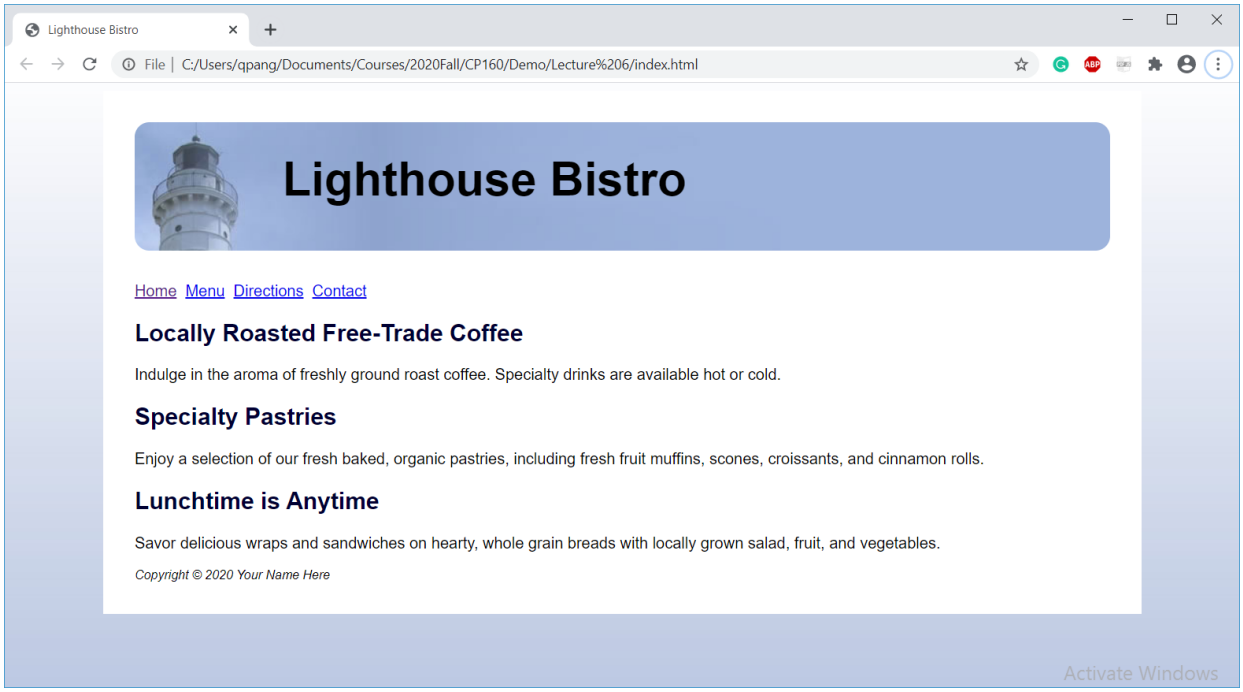

Figure 3 The web page is centered with CSS.

Hands-On Practice 3: Configure opacity

In this Hands-On Practice, you'll work with the opacity property as you configure the web page shown in **Figure 4**.

1. Create a new folder called **opacitych6**. Copy fall.jpg file from the **starters** folder to your opacitych6 folder. Open the template.html file in a text editor. Save it in your **opacitych6** folder with the name index.html. Change the page title to "Fall Nature Hikes".
2. Let's create the structure of the web page with a **div** that contains an **h1** element. Add the following code to your web page in the body section:

```
<div id="content">
<h1> Fall Nature Hikes</h1>
</div>
```

3. Now, add style tags to the head section and configure the embedded CSS. You'll create an id named `content` to display the fall.jpg as a background image that does not repeat. The `content` id also has a width of 640 pixels, a height of 480 pixels, auto margins (which will center the object horizontally in the browser viewport), and 20 pixels of top padding. The code follows:

```
#content { background-image: url(fall.jpg);
           background-repeat: no-repeat;
           margin: auto;
           width: 640px;
           height: 480px;
           padding-top: 20px; }
```

4. Now configure the h1 selector to have a white background color, opacity set to 0.6, font size set to 4em, 10 pixels of padding, and a 40-pixel left margin. Sample code follows.

```
h1 { background-color: #FFFFFF;
     opacity: 0.6;
     font-size: 4em;
     padding: 10px;
     margin-left: 40px; }
```

Save the file. When you test your index.html file in a browser that supports opacity, the display should look similar to the page shown in **Figure 4**.

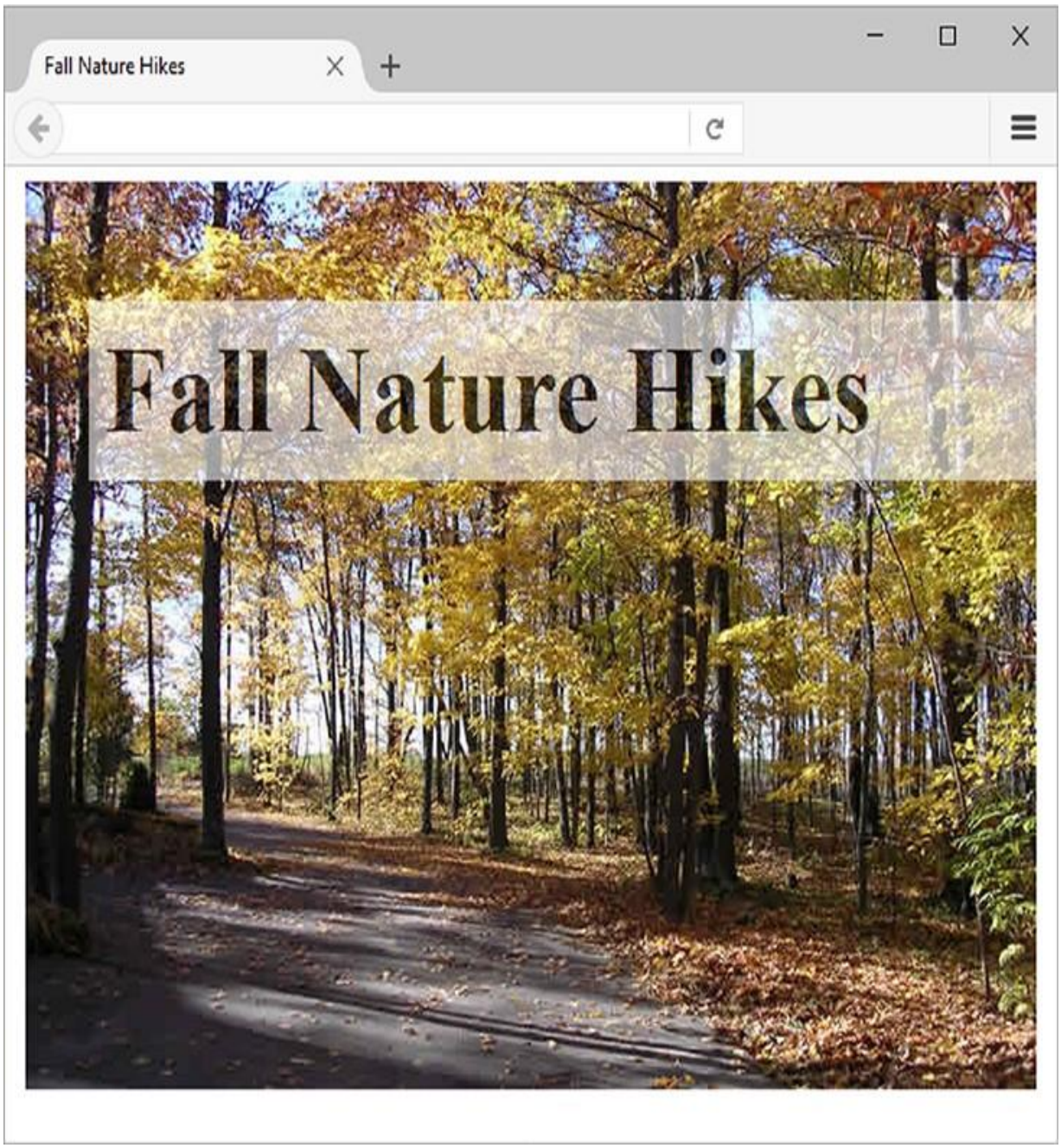

Figure 4 The background and text of the h1 area is transparent.

Hands-On Practice 4: Configure CSS gradient backgrounds

You'll work with CSS gradient backgrounds in this Hands-On Practice. Create a new folder called **gradientch6**. Copy the **starter4.html** file from starters folder into your gradientch6 folder. Rename the file **index.html**.

Launch a text editor and open the file.

1. First, you will configure a linear gradient. Code embedded CSS in the head section. Configure the body of the web page to display a fallback orchid background color of #DA70D6, and a linear gradient background that blends white to orchid from top to bottom without repeating:

```
body {  background-color: #DA70D6;
        background-image: linear-gradient(to bottom, #FFFFFF,
#DA70D6);
        background-repeat: no-repeat; }
```

2. Save your file and test it in a modern browser. The display should be similar to the results shown in **Figure 5**. The background gradient displays behind the page content, so scroll down the page to see the full gradient.
3. Next, you will configure a radial gradient. Edit the body section of the web page and code change the text within the h1 element to: Radial Gradient.
4. Edit the CSS and modify the value of the `background-image` property to configure a radial gradient linear gradient that blends white to orchid from center outward without repeating:

```
body { background-color: #DA70D6;
       background-image: radial-gradient(#FFFFFF, #DA70D6);
       background-repeat: no-repeat; }
```

5. Save your file and test it in a modern browser. The display should be similar to the results shown in **Figure 6**. Scroll down the page to see the full gradient.

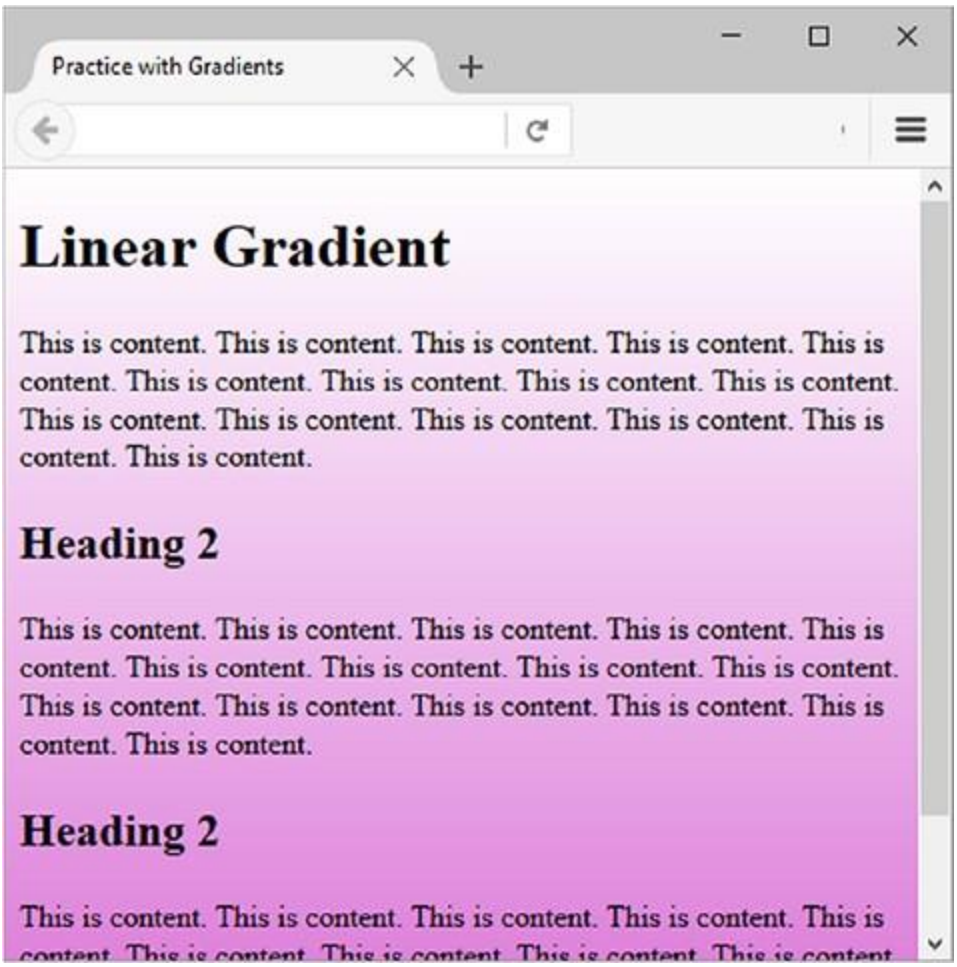

Figure 5 Linear gradient background.

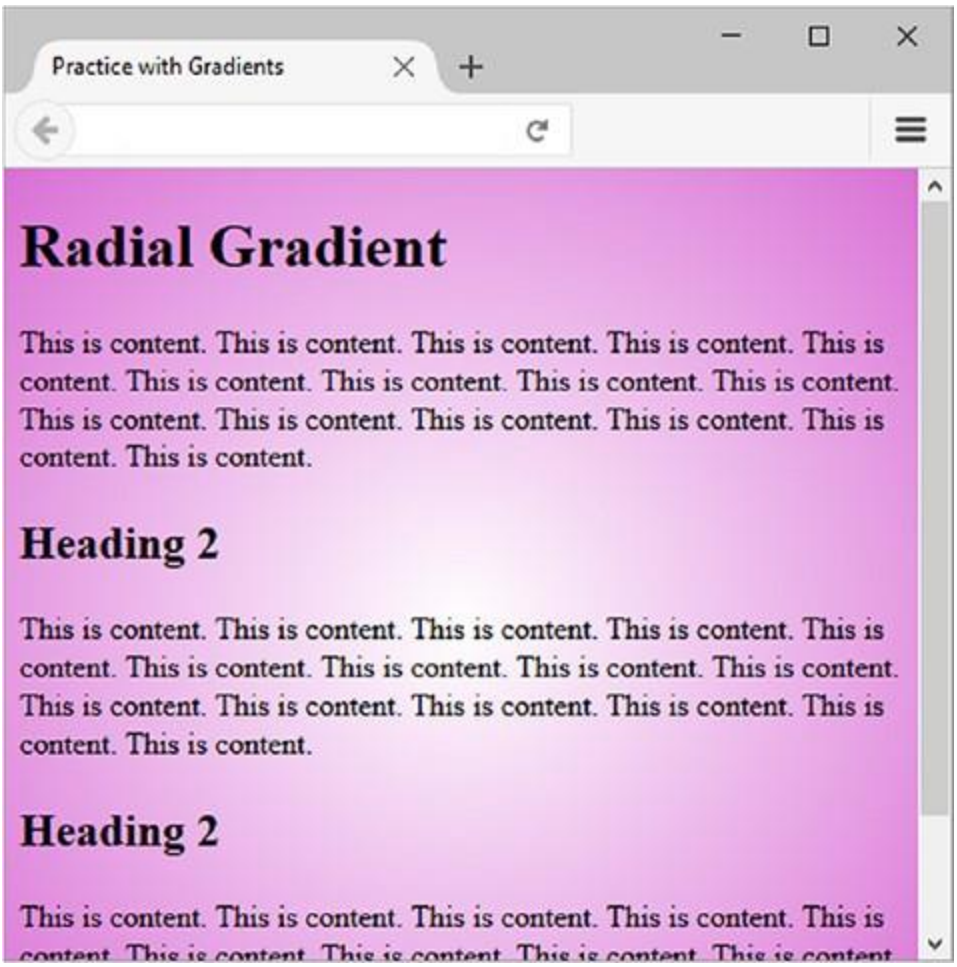

Figure 6 Radial gradient background.

Lab Completion / Submission:

Complete all the lab practices. Take the screenshots of your completed webpages; put them into a single word file and submit it to **Blackboard -> CP160 -> Assessments -> Lab 6 / Assignment 6**; due date: today

After-lab Assignment:

1. Create a one-paragraph conclusion of what you have learned during the lab today.
2. Write the CSS to configure an id named **content** with 70% opacity
3. Write the CSS for an id named notice that is configured with width set to 80% and centered.
4. Write the CSS to configure the body element with a linear gradient background that blends from black to medium blue.

Submit to **Blackboard -> CP160 -> Assessments -> Lab 6 / Assignment 6;** due date: 1 week from today.

***Project** (Please refer to Lab 3 and 4 manuals for more instructions; Due date: Oct 21, submit to **Blackboard -> CP160 -> Assessments -> Project Proposal**)

- Web Project Topic Selection
- Web Project Planning Analysis
- Web Project Site Map
- Web Project Page Layout Design